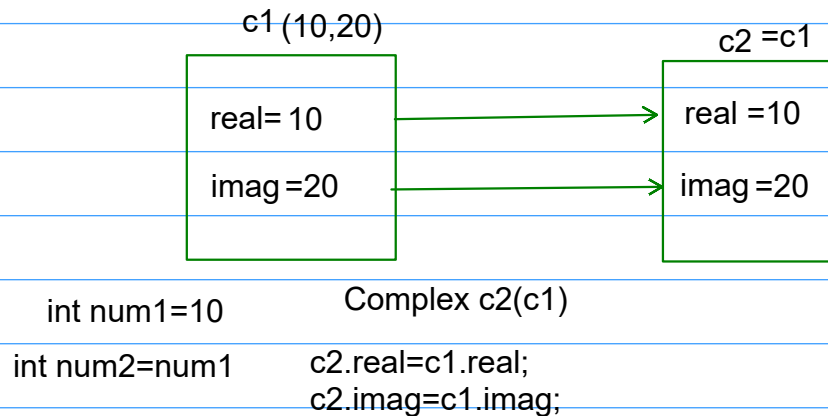


Complex Number

1 real and 1 img.

$$\begin{array}{ccc} \underline{a} + i \underline{b} & & \text{real} \\ \text{real} & \text{img} & \text{img} \end{array}$$



$c_1(10, 20)$ $c_2(20, 30)$
 $c_1 + c_2 \Rightarrow c_1.sum(c_2)$
 $10 + i 20$
 $20 + i 30$ $c_1.sum(c_2)$
 $\checkmark 40 + i 50$ Complex number
 Complex sum(Complex c2)
 { Complex temp;
 temp.real = this->real + c2.real
 temp.imag = this->img + c2.img
 return temp;

c1.sum

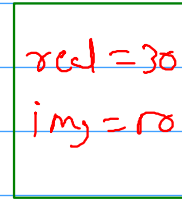
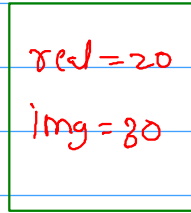
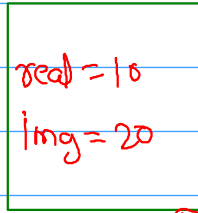
main

c1(10,20)

c2(20,30)

c3

completing (39,50)



=

copy

3000

4000

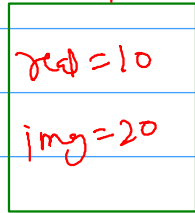
c3

this

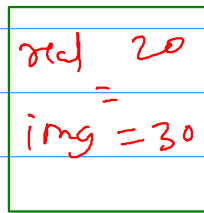
©

anonymous

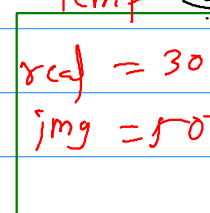
sum



+

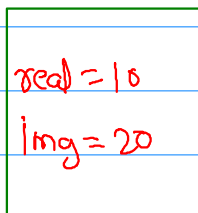


=

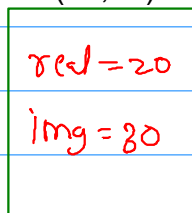


other

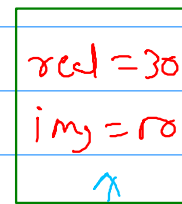
(39,50)



c2(20,30)



=



c3

copy

anonymous

3000

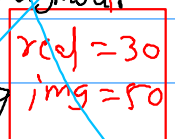
4000

this

©

temp

copy



real=10

img=20

+

real=20

img=30

=

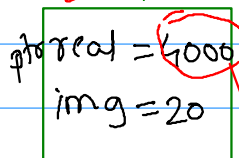
real=30

img=50

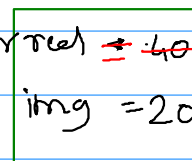
stack

other

c1



c2 = c1



NULL

destructor

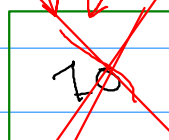
delete ptrReal

ptrReal=NULL

ptrReal = new int;

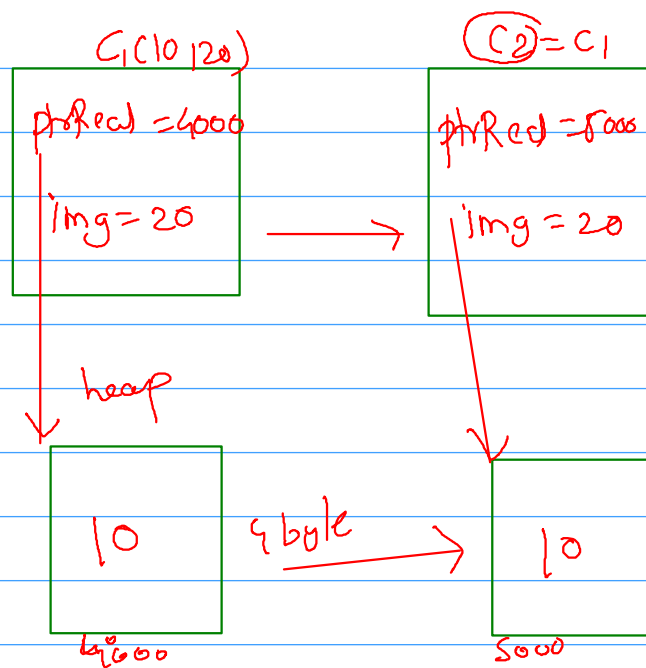
c1(10,20)

heap *ptrReal = real



4 byte

4000



tightly coupled Association $\Rightarrow$ has-a loosely coupled
 composition Aggregation
 human has a heart
Car has a engine
 Employee has a hiredale
 Person has a DoB

Car has a Engine

dependent
Car

dependency
Engine

```

    dependency
class Engine
{
    int cc;
    int fuel;
}

    dependent
class car
{
    int price;
    Engine e; // data member
}
  
```

```

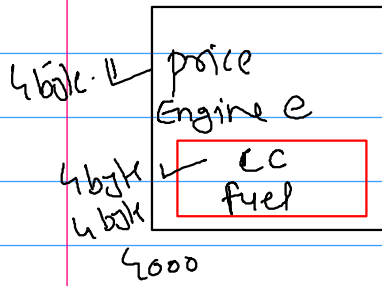
int main()
{
    Car C (1000, 100, 100)
  }
  
```

```

public: Car (price, fuel, cc) : Engine (fuel, cc)
{
}
  
```

Car
int price
Engine e

Car C₁



sizeof(C₁)

⇒ 12 byte

size(engine) ⇒ 8 byte.

tightly coupled

Person has a DOB.

~~dependent~~

dependency

Class Date

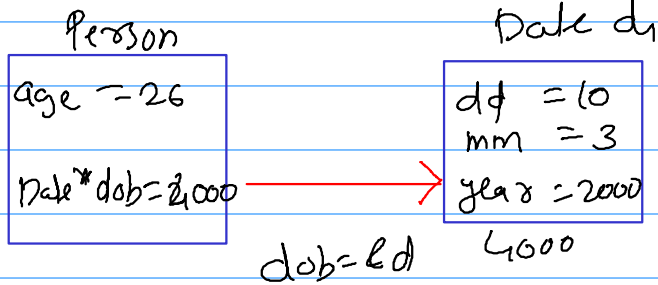
```
{ int day,  
  int month,  
  int year,  
}
```

Class Person

```
{ int age  
  Date * Dob
```

int main()

```
  Date d1(10, 3, 2000)  
  Person(26, d1);
```



Inheritance

if one class acquires properties of another class $\Rightarrow$ here inheritance.

is-a $\Rightarrow$ inheritance.

Employee is a Person
SavingsAccount is a Account.
Car is a Vehicle

Child B inherited properties from parent

Class parent

```
{  
    int num1;  
    void f1() { }  
};
```

Class child : public Parent

Class Name

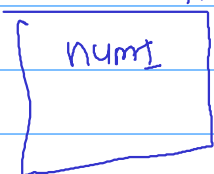
mode of inheritance

```
{  
    int num2;
```

```
    void f2() { }  
};
```

int num1;
void f1();

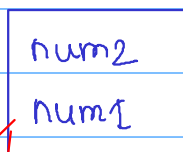
Parent P1



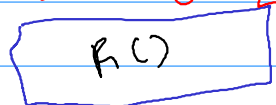
Child C1

sizeof(C1)

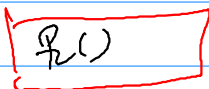
8 byte



Code Segment



Code seg



Size of child class

= non-static data member of Parent
+ non-static data member of child

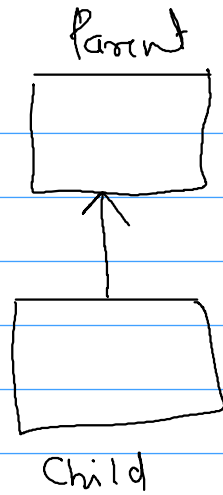
main()

Parent P1;

P1.f1()
P1.f2();

Child C1;
C1.f2() ✓
C1.f1()

single inheritance



public mode of inheritance

	class	derived	indirect	other	friend
public	✓	✓	✓		✓
private	✓	✗	✗		✓
protected	✓	✓	✗		✓

private

public/protected becomes private

